

Yaowen Shi

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EDUCATION

South China University of Technology (SCUT)

B.S. in Big Data Management and Application

Sep. 2023 – Jun. 2027 (Expected)

- Academic standing: GPA: **3.95/4.00**; average score: **93.7**; major rank: **3/161**
- Selected courses: Calculus I and II (**99, 100**); Linear Algebra and Analytic Geometry (**100**); Operations Research (**99**); Python Data Analysis (**97**); Probability Theory and Mathematical Statistics (**96**); Large Language Models and GenAI (**96**); Econometrics (**95**); Data Structures (**95**); ... **Ranked 1st in 11 courses.**

RESEARCH INTERESTS

I am interested in **multimodal agent systems** that connect reasoning, planning, memory, tool use, and multimodal interaction with real creation workflows. My research focuses on **LLM-based agents for visual and multimodal generation**: agents that understand human intent, decompose open-ended tasks, coordinate specialized models or tools, and improve outputs through **planning, feedback, and self-reflection**. I am broadly interested in image generation, video generation, 3D/4D generation, visual reasoning, and autonomous task execution, with the long-term goal of making foundation models more capable, controllable, and useful for complex creative tasks.

PUBLICATIONS

SymGround: Visual Program Synthesis with Symbol-to-Vision Grounding

Yaowen Shi, Jinxiu Liu

Under Review at EMNLP 2026.

- Identified visual-to-program credit assignment as the bottleneck in executable visual synthesis, and built a relation-grounded loop that maps rendered residuals to editable program symbols before each localized edit.
- Evaluated on SlideBench, BlenderGym, and BlenderBench, outperforming VIGA while reducing token usage and invalid executions.

EXPERIENCE

Artist-Created 3D Mesh Generation

Mar. 2025 – Apr. 2025

Collaborator: Jinxiu Liu, Research Scientist in Nex-AGI

- Studied efficient mesh compression and tokenization for autoregressive artist-created mesh generation: explored VQ-VAE-based mesh vocabularies and adjacency-aware tokenization to shorten mesh token sequences, increase the number of faces models can generate, and preserve geometric structure; also investigated human-preference alignment methods such as RLHF/DPO to improve perceptual quality and user preference.

Image Generation

Apr. 2025 – Jun. 2025

Collaborator: Jinxiu Liu, Research Scientist in Nex-AGI

- Prototyped self-reflection and reinforcement learning for flow matching text-to-image models (e.g., FLUX): turned intermediate samples into self-critique signals to define rewards, and learned an adaptive conditioning/guidance schedule under a fixed sampling budget; improved prompt adherence, text rendering, and subjective preference while reducing trajectory drift.

Controllable Video Generation

Jul. 2025 – Aug. 2025

Collaborator: Jinxiu Liu, Research Scientist in Nex-AGI

- Studied training-free video motion transfer methods with DiT-based video models such as Wan 2.1 and CogVideoX (5B/2B), using DiTFlow as a baseline: extracted patch-wise motion cues from source videos via cross-frame attention / Attention Motion Flow (AMF), applied inference-time motion losses to latent denoising or positional embeddings, and evaluated motion transfer quality and temporal consistency under subject or scene changes.

Omni Multimodal Generation Agent System

Sep. 2025 – Jan. 2026

Collaborator: Jinxiu Liu, Research Scientist in Nex-AGI

- Built an omni multimodal generation system with NexAU as the agent-universe orchestration layer and ComfyUI as the workflow execution backend: defined tool-using agents, skills, and workflow primitives to plan and execute image/video generation, image/video editing, and related multimodal creation tasks; inspired by omni systems such as Nano Banana and BAGEL, and achieved state-of-the-art on ComfyBench.

Vision-as-Inverse-Graphics Program Synthesis

Feb. 2026 – May 2026

Collaborator: Jinxiu Liu, Research Scientist in Nex-AGI

- Designed a relation-grounded visual program synthesis pipeline for executable slides and Blender scenes: separated visual attribution from evidence-guided editing, represented each revision as a testable symbol-to-vision hypothesis, and accumulated edit-render memory to avoid unsupported retries and broad rewrites.
- Benchmarked SymGround on SlideBench, BlenderGym, and BlenderBench against VIGA, improving aggregate performance while reducing token usage and invalid-execution rate.

Benchmarking Agents for 3D Scene Generation

May 2026 – Present

Advised by Yandong Wen, Assistant Professor at Westlake University

- Built a benchmark for evaluating agent capabilities in generating Blender 3D scenes, focusing on instruction following, spatial composition, executable scene construction, and visual fidelity.
- Designed task cases and evaluation criteria to compare how agent systems plan, implement, render, and revise complex 3D scenes in Blender.

PROJECTS

OpenDinq | *Evidence-backed AI-native profiles, card workspaces, and explainable people discovery* GitHub

- Built an open-source product alpha for evidence-backed AI-native profiles: source ingestion from GitHub, websites, OpenAlex, arXiv, ORCID, manual links, and notes, then normalized, deduped, and quality-scored claims.
- Implemented card workspaces for profile review and curation, public profile publishing, and explainable people discovery with matched claims, artifacts, evidence snippets, and score breakdowns.

AWARDS

Meritorious Winner (Top 7%), Mathematical Contest in Modeling (MCM) / Interdisciplinary Contest in Modeling (ICM) 2025

Second Prize (Top 15%), Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM), Guangdong Division 2024

TECHNICAL SKILLS

Languages: C++, Python, SQL, JavaScript, HTML, CSS

ML: PyTorch, Transformers, diffusers

Developer Tools: Git, Docker

English: CET-6